T7.2.3-Worksheet 2-Pseudocode-Count-Controlled Loop

Answer the following questions.

1. Write pseudocode using a `FOR` loop that outputs the message `"Welcome to Computer Science"` exactly 10 times.

   Answer:

```
DECLARE Count : INTEGER

FOR Count ← 1 TO 10
    OUTPUT "Welcome to Computer Science"
NEXT Count
```

2. Write pseudocode using a `FOR` loop that asks the user to enter 5 numbers one by one, calculates the total sum of these 5 numbers, and outputs the final total.

   Answer:

```
DECLARE Total : REAL
DECLARE Number : REAL
DECLARE Counter : INTEGER

Total ← 0

FOR Counter ← 1 TO 5
    OUTPUT "Enter a number: "
    INPUT Number
    Total ← Total + Number
NEXT Counter

OUTPUT "The total sum is: ", Total
```

3. Write pseudocode to input the test scores for a fixed group of exactly 8 students. For each score entered, determine if it is 50 or above, and if so, record it as a pass. Once all 8 scores have been processed, output the final number of students who passed. Declare all variables used.

   Answer:

```
DECLARE PassCount : INTEGER
DECLARE Index : INTEGER
DECLARE Score : REAL

PassCount ← 0

FOR Index ← 1 TO 8
    OUTPUT "Enter student score: "
    INPUT Score
    IF Score >= 50 THEN
        PassCount ← PassCount + 1
    ENDIF
NEXT Index

OUTPUT "Total number of students who passed: ", PassCount
```

4. Write pseudocode that prompts the user to enter a single whole number. Using a `FOR` loop, output the multiplication table for that number from 1 to 12 (e.g., displaying `Number * 1, Number * 2`, up to `Number * 12`).

   Answer:

```
DECLARE BaseNumber : INTEGER
DECLARE Multiplier : INTEGER
DECLARE Result : INTEGER

OUTPUT "Enter a number for its multiplication table: "
INPUT BaseNumber

FOR Multiplier ← 1 TO 12
    Result ← BaseNumber * Multiplier
    OUTPUT BaseNumber, " x ", Multiplier, " = ", Result
NEXT Multiplier
```

5. A weather station records the daily temperature in Celsius at 12:00 PM every day for a week (7 days).

   Write pseudocode using a `FOR` loop that:

   a. Inputs the daily temperature for each of the 7 days.
   b. Calculates the total sum and the average temperature for the week.
   c. Determines and stores the highest recorded temperature of the week.
   d. Outputs the calculated average temperature and the highest temperature.

   Declare all variables used with appropriate data types.

   Answer:

```
DECLARE Day : INTEGER
DECLARE Temperature : REAL
DECLARE TotalTemp : REAL
DECLARE AverageTemp : REAL
DECLARE HighestTemp : REAL

TotalTemp ← 0
HighestTemp ← -100.0

FOR Day ← 1 TO 7
    OUTPUT "Enter temperature for Day ", Day, ": "
    INPUT Temperature
    TotalTemp ← TotalTemp + Temperature
    IF Temperature > HighestTemp THEN
        HighestTemp ← Temperature
    ENDIF
NEXT Day

AverageTemp ← TotalTemp / 7

OUTPUT "Average temperature for the week: ", AverageTemp
OUTPUT "Highest recorded temperature: ", HighestTemp
```

6. A bookstore processes orders for a fixed batch of exactly 15 customers. Each customer enters the total cost of their book purchase.

Write an algorithm in pseudocode to:

a. Input the order amount for each of the 15 customers.
b. Calculate the total revenue generated from all 15 orders.
c. Count how many customers qualify for free gift wrapping (orders of **$50 or more**).
d. Count how many customers placed small orders under **$20**.
e. Output the total revenue, the number of qualifying gift wrap orders, and the number of small orders after all 15 customer purchases are entered.

Declare all variables used with appropriate data types.

Answer:

```
DECLARE CustomerCount : INTEGER
DECLARE OrderAmount : REAL
DECLARE TotalRevenue : REAL
DECLARE GiftWrapCount : INTEGER
DECLARE SmallOrderCount : INTEGER

TotalRevenue ← 0
GiftWrapCount ← 0
SmallOrderCount ← 0

FOR CustomerCount ← 1 TO 15
    OUTPUT "Enter order amount for customer ", CustomerCount, ": $"
    INPUT OrderAmount
    TotalRevenue ← TotalRevenue + OrderAmount

    IF OrderAmount >= 50.0 THEN
        GiftWrapCount ← GiftWrapCount + 1
    ELSE
        IF OrderAmount < 20.0 THEN
            SmallOrderCount ← SmallOrderCount + 1
        ENDIF
    ENDIF
NEXT CustomerCount

OUTPUT "Total Revenue: $", TotalRevenue
OUTPUT "Orders eligible for free gift wrap: ", GiftWrapCount
OUTPUT "Small orders under $20: ", SmallOrderCount
```

7. Challenge Yourself Question:

A construction plant manufactures a pre-ordered batch of exactly 20 solid concrete pillars shaped as right trapezoidal prisms. Write an algorithm in pseudocode that inputs the two parallel bases (A and B), the perpendicular height (H), and the length (L) in meters for each of the 20 pillars, validating that all entered measurements are greater than zero before computing the volume using $Volume = \frac{A+B}{2} \times H \times L$. The program must calculate the total concrete volume required for the batch, track the smallest pillar volume produced, count how many individual pillars exceed a standard heavy-duty threshold of $15.0 m^3$, and output these three final values once all 20 pillars have been processed. Declare all variables used with standard IGCSE data types.

Answer:

```
DECLARE PillarCount : INTEGER
DECLARE BaseA : REAL
DECLARE BaseB : REAL
DECLARE Height : REAL
DECLARE Length : REAL
DECLARE Volume : REAL
DECLARE TotalVolume : REAL
DECLARE MinVolume : REAL
DECLARE HeavyDutyCount : REAL

TotalVolume ← 0.0
MinVolume ← 999999.0
HeavyDutyCount ← 0

FOR PillarCount ← 1 TO 20
     OUTPUT "Enter Base A, Base B, Height, and Length for pillar ",
PillarCount, ":"
    INPUT BaseA, BaseB, Height, Length

     IF BaseA > 0.0 AND BaseB > 0.0 AND Height > 0.0 AND Length > 0.0
THEN
        Volume ← ((BaseA + BaseB) / 2.0) * Height * Length
        TotalVolume ← TotalVolume + Volume

        IF Volume < MinVolume THEN
            MinVolume ← Volume
        ENDIF

        IF Volume > 15.0 THEN
            HeavyDutyCount ← HeavyDutyCount + 1
```

```
        ENDIF
    ELSE
        OUTPUT "Invalid dimensions entered for pillar ", PillarCount
    ENDIF
NEXT PillarCount

OUTPUT "Total concrete volume required: ", TotalVolume
OUTPUT "Smallest pillar volume: ", MinVolume
OUTPUT "Number of heavy-duty pillars (>15 m^3): ", HeavyDutyCount
```

End of Question Paper